

ITA0606 – Machine Learning

Exp-18

Program:

Perceptron based IRIS Classification

Import required libraries

```
from sklearn.datasets import load_iris
```

```
from sklearn.model_selection import train_test_split
```

```
from sklearn.linear_model import Perceptron
```

```
from sklearn.metrics import accuracy_score, confusion_matrix
```

Step 1: Load the Iris dataset

```
iris = load_iris()
```

```
X = iris.data
```

```
y = iris.target
```

Step 2: Split the dataset into training and testing data

```
X_train, X_test, y_train, y_test = train_test_split(
```

```
    X, y,
```

```
    test_size=0.3,
```

```
    random_state=42
```

```
)
```

Step 3: Create Perceptron model

```
model = Perceptron(
```

```
    max_iter=1000,
```

```
    eta0=0.1,  
    random_state=42  
)
```

```
# Step 4: Train the model
```

```
model.fit(X_train, y_train)
```

```
# Step 5: Predict the test data
```

```
y_pred = model.predict(X_test)
```

```
# Step 6: Display actual and predicted values
```

```
print("Actual Classes:")
```

```
print(y_test)
```

```
print("\nPredicted Classes:")
```

```
print(y_pred)
```

```
# Step 7: Calculate accuracy
```

```
accuracy = accuracy_score(y_test, y_pred)
```

```
print("\nAccuracy:")
```

```
print(accuracy * 100, "%")
```

```
# Step 8: Display confusion matrix
```

```
cm = confusion_matrix(y_test, y_pred)
```

```
print("\nConfusion Matrix:")
```

```
print(cm)
```

Step 9: Display class names

```
print("\nClass Names:")
```

```
print(iris.target_names)
```

Step 10: Predict a new Iris flower

```
new_flower = [[5.1, 3.5, 1.4, 0.2]]
```

```
prediction = model.predict(new_flower)
```

```
print("\nNew Flower Features:")
```

```
print(new_flower)
```

```
print("\nPredicted Class:")
```

```
print(iris.target_names[prediction[0]])
```

OUTPUT:

```
*** Actual Classes:
[1 0 2 1 1 0 1 2 1 1 2 0 0 0 0 1 2 1 1 2 0 2 0 2 2 2 2 2 0 0 0 0 1 0 0 2 1
 0 0 0 2 1 1 0 0]

Predicted Classes:
[0 0 2 0 0 0 0 0 1 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0
 0 0 0 0 0 0 0 0]

Accuracy:
46.666666666666664 %

Confusion Matrix:
[[19  0  0]
 [12  1  0]
 [11  1  1]]

Class Names:
['setosa' 'versicolor' 'virginica']

New Flower Features:
[[5.1, 3.5, 1.4, 0.2]]

Predicted Class:
setosa
```